



E0832

Place student sticker here

Note:

- During the attendance check a sticker containing a unique code will be put on this exam.
- This code contains a unique number that associates this exam with your registration number.
- This number is printed both next to the code and to the signature field in the attendance check list.

Machine Learning

Exam: IN2064 / Endterm

Date: Thursday 12th February, 2026

Examiner: Prof. Dr. Stephan Günnemann

Time: 08:00 – 10:00

P 1

P 2

P 3

P 4

P 5

P 6

P 7

P 8

P 9

--	--	--	--	--	--	--	--	--

Working instructions

- This exam consists of **16 pages** with a total of **9 problems**.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 44 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
 - Two-sided DIN A4 sheet of handwritten notes (a print of digitally handwritten notes is allowed).
- **No other material (e.g. books, cell phones, calculators) is allowed!**
- Physically turn off all electronic devices, put them into your bag and close the bag.
- There is scratch paper at the end of the exam.
- Write your answers only in the provided solution boxes or the scratch paper.
- If you solve a task on the scratch paper, clearly reference it in the main solution box.
- All sheets (including scratch paper) have to be returned at the end.
- **Only use a black or a blue pen (no pencils, red or greens pens!)**
- **For problems that say “Justify your answer” you only get points if you provide a valid explanation.**
- **For problems that say “Derive” you only get points if you provide a valid mathematical derivation.**
- **For problems that say “Prove” you only get points if you provide a valid mathematical proof.**
- If a problem does not say “Justify your answer”, “Derive” or “Prove”, it is sufficient to only provide the correct answer.

Left room from _____ to _____ / Early submission at _____



☐ Exam empty





CIT-IN2064-01-20260212
4794-01-00008-F0832-02





Problem 2 Probabilistic Inference (6 credits)

Let $X \in \{0, 1\}$ be a random variable denoting the result of a (possibly biased) coin flip ($X = 0$ for tails and $X = 1$ for heads). Let θ_1 denote the probability of observing heads. Suppose that you have a smart device that looks at the flipped coin and says its outcome, which we denote as Y . However, this device is not always reliable and only says the correct outcome with probability θ_2 . Formally, we have

$$p(y|x, \theta_2) = \begin{cases} \theta_2, & \text{if } y = x \\ 1 - \theta_2, & \text{otherwise.} \end{cases}$$

We assume that θ_2 is independent of θ_1 .

a) Write down the joint probability distribution $p(x, y|\theta)$ in terms of the parameter vector $\theta = (\theta_1, \theta_2)^T$ for all possible values of x and y .

0
1





- 0 ☐
- 1 ☐
- 2 ☐

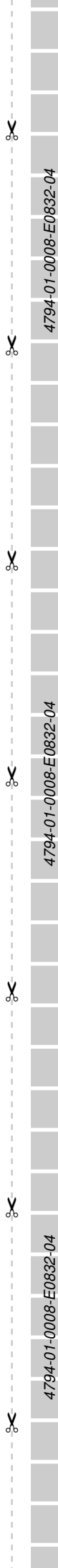
b) Suppose you collect the following dataset of (x, y) pairs:

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n = \{(1, 1), (1, 0), (0, 0), (0, 1), (1, 1), (0, 0), (0, 1)\}.$$

Derive the maximum likelihood estimates (MLEs) $\hat{\theta}_1$ and $\hat{\theta}_2$ for the 2 parameters θ_1 and θ_2 .

- 0 ☐
- 1 ☐

c) Now instead of the previous 2-parameter model, consider a model with 4 parameters $\theta = (\theta_{0,0}, \theta_{0,1}, \theta_{1,0}, \theta_{1,1})$ representing $p(x, y|\theta) = \theta_{x,y}$, where $\theta_{0,0} + \theta_{0,1} + \theta_{1,0} + \theta_{1,1} = 1$. Write down the likelihood function of this model on the observed dataset $\mathcal{D}$.





d) Suppose you are not sure which of the two models to use. So, you decide to compute the leave-one-out cross validated log likelihood of both models as follows: $L(m) = \sum_{i=1}^n \log p(x_i, y_i | m, \hat{\theta}(\mathcal{D}_{-i}))$, where m selects one of the two models and $\hat{\theta}(\mathcal{D}_{-i})$ denotes the MLE of the model parameters computed on $\mathcal{D}$ excluding the i -th pair. Which model is preferred according to this metric? **Justify** your answer.

Hint 1: For the 4-parameter model, the MLEs are given as $\theta_{x,y} = \frac{N_{(x,y)}}{N}$, where $N_{(x,y)}$ is the number of pairs in $\mathcal{D}$ having the values (x, y) and N is the total size of $\mathcal{D}$.

Hint 2: Specifically consider the term where the second training sample is left out.





Problem 3 Poisson Regression (4 credits)

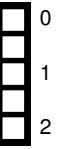
In this task, we consider a regression problem where the target variable is a count variable, e.g., the number of emails expected on a specific day. In this case, a common choice for the likelihood function is the Poisson distribution, which has its support over positive integers. Specifically, a random variable $Y \in \mathbb{N}_0$ has a Poisson distribution with rate $\lambda > 0$ if its probability mass function is given by:

$$p(y|\lambda) = \frac{e^{-\lambda} \lambda^y}{y!}.$$

In Poisson regression, instead of assuming a fixed λ , we model it as a function of some input data $\mathbf{x} \in \mathbb{R}^d$, e.g., the day of the week might influence the number of expected emails on that day. Specifically, we assume $\log \lambda = \boldsymbol{\theta}^\top \mathbf{x}$, where $\boldsymbol{\theta} \in \mathbb{R}^d$ is the parameter vector.

- 0 ☐ a) Consider a training dataset consisting of features $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_n)^\top$ with respective count targets $\mathbf{y} = (y_1, \dots, y_n)^\top$, where $\mathbf{x}_i \in \mathbb{R}^d$ and $y_i \in \mathbb{N}_0$. Write down the negative log-likelihood, $-\log p(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta})$, of a Poisson regression model with parameter $\boldsymbol{\theta} \in \mathbb{R}^d$ in terms of $\boldsymbol{\theta}$, $\mathbf{x}_i$ and y_i .
- 1 ☐





b) To find the optimal parameters θ^* , we need to minimize the negative log-likelihood. However, this optimization problem has no closed-form solution, so we need to use gradient-based methods instead. **Derive** the gradient descent update rule for the j -th parameter θ_j of a Poisson regression model. Assume a learning rate $\alpha \in \mathbb{R}$.

c) What are the advantages of using a Poisson regression model for a count prediction problem over (i) a linear regression model, and (ii) a linear classification model? Name one advantage for each case.





Problem 4 Linear Classification (4 credits)

Consider a binary classification problem on the feature space $x \in \mathbb{N}_0$ of natural numbers (including 0) with labels $y \in \{0, 1\}$. You are given a probabilistic generative classifier that models the prior class probabilities as

$$p(y = 0) = \frac{4}{5} \quad \text{and} \quad p(y = 1) = \frac{1}{5},$$

and the class-conditional feature distribution as a Poisson distributions, i.e.,

$$p(x \mid y = k) = \frac{e^{-\lambda_k} \lambda_k^x}{x!}$$

for $k \in \{0, 1\}$, $\lambda_0 \neq \lambda_1$, and $\lambda_i > 0$.

0 ☐

1 ☐

2 ☐

a) Classify the point $x = 2$ given $\lambda_0 = 1$ and $\lambda_1 = 2$. **Justify your answer!**

0 ☐

1 ☐

2 ☐

b) Derive the decision boundary of this classifier in terms of λ_0 and λ_1 . **Justify your answer!**





Problem 5 Optimization (6 credits)

Recall that the negative log-likelihood (NLL) of a logistic regression model can be written as

$$E(\mathbf{w}) = -\log p(\mathbf{y}|\mathbf{X}, \mathbf{w}).$$

Definition: A differentiable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is called **strongly convex** if

$$f(\mathbf{y}) - f(\mathbf{x}) \geq (\mathbf{y} - \mathbf{x})^T \nabla_{\mathbf{x}} f(\mathbf{x}) + \frac{\mu}{2} \|\mathbf{y} - \mathbf{x}\|^2$$

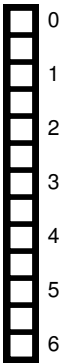
for a parameter $\mu > 0$ and all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$.

Note that for $\mu = 0$, this definition reduces to the definition of convexity as in the lecture.

Interestingly, one can show that strongly convex functions have a unique global minimum. We want to use this to show that the NLL of a logistic regression model with L2-regularization has a unique global minimum.

Prove that the function $h(\mathbf{w}) = E(\mathbf{w}) + \frac{\lambda}{2} \|\mathbf{w}\|_2^2$ is strongly convex for any $\lambda > 0$.

Hint: You may use without proof that the negative log-likelihood $E(\mathbf{w})$ is a convex function on $\mathbb{R}^d$.





Problem 6 Deep Learning (6 credits)

Consider a fully connected neural network with one hidden layer of K neurons. The network as a whole is defined by this function:

$$f(\mathbf{x}) = \sum_{k=1}^K v_k \text{ReLU}(\mathbf{W}_k^\top \mathbf{x} + b_k),$$

where $\mathbf{x} \in \mathbb{R}^D$ is the input, $v_k, b_k \in \mathbb{R}$, and $\mathbf{W} \in \mathbb{R}^{K \times D}$ is the weight matrix of the hidden layer.

$\mathbf{W}_k \in \mathbb{R}^{1 \times D}$ denotes the k -th row of $\mathbf{W}$.

The activation function is:

$$\text{ReLU}(t) = \max(0, t), \quad \text{ReLU}'(0) = 0.$$

0

1

2

3

a) **Derive** the partial derivative $\frac{\partial f(\mathbf{x})}{\partial x_d}$, for $d \in \{1, \dots, D\}$.

0

1

2

b) How many distinct values can the gradient $\nabla_{\mathbf{x}} f(\mathbf{x}) \in \mathbb{R}^D$ take for $\mathbf{x} \in \mathbb{R}^D$? **Justify your answer!**

0

1

c) How many distinct values can the gradient $\nabla_{\mathbf{x}} f(\mathbf{x}) \in \mathbb{R}^D$ take if we replace the ReLU activation function by $\text{LeakyReLU}(t) = \max(\alpha t, t)$, $\alpha > 0$?
We assume that $\text{LeakyReLU}'(0) = \alpha$.





Problem 7 Support Vector Machines (7 credits)

Consider the soft-margin SVM primal problem:

$$\min_{\mathbf{w}, b, \xi} \quad \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^N \xi_i \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0 \quad \forall i.$$

a) Write down the dual optimization problem. What are the constraints on the Lagrange multipliers α_i ?

☐	0
☐	1

b) Given an optimal solution α^* , how do you recover $\mathbf{w}^*$? For which support vectors can you compute b^* , and why? **Justify your answer!**

☐	0
☐	1





0	☐
1	☐
2	☐
3	☐
4	☐
5	☐

c) Consider the polynomial kernel $k(\mathbf{x}_1, \mathbf{x}_2) = (\mathbf{x}_1^\top \mathbf{x}_2 + 1)^2$ for $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^2$. **Derive** an explicit feature mapping $\phi : \mathbb{R}^2 \rightarrow \mathbb{R}^M$ such that $k(\mathbf{x}_1, \mathbf{x}_2) = \phi(\mathbf{x}_1)^\top \phi(\mathbf{x}_2)$.



4794-01-0008-E0832-12

4794-01-0008-E0832-12

4794-01-0008-E0832-12





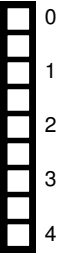
Problem 8 Dimensionality Reduction (4 credits)

Let $\mathbf{X} \in \mathbb{R}^{n \times d}$ be a data matrix of n samples with d features and $\mathbf{\Sigma} \in \mathbb{R}^{n \times n}$ its empirical covariance matrix. We define the *total variance* of $\mathbf{X}$ as

$$\text{TVar}(\mathbf{X}) = \text{Tr}(\mathbf{\Sigma}) = \sum_{i=1}^n \Sigma_{ii}.$$

Let $\mathbf{\Gamma} \in \mathbb{R}^{d \times d}$ be an alternative basis of $\mathbb{R}^d$, i.e. its columns are pairwise orthogonal and of unit norm. **Prove** that the total variance of $\mathbf{X}$ does not change if we project our data onto the basis $\mathbf{\Gamma}$.

Hint: You can cycle matrices inside the trace operator, i.e. $\text{Tr}(\mathbf{ABC}) = \text{Tr}(\mathbf{BCA}) = \text{Tr}(\mathbf{CAB})$.



4794-01-0008-E0832-13

4794-01-0008-E0832-13

4794-01-0008-E0832-13



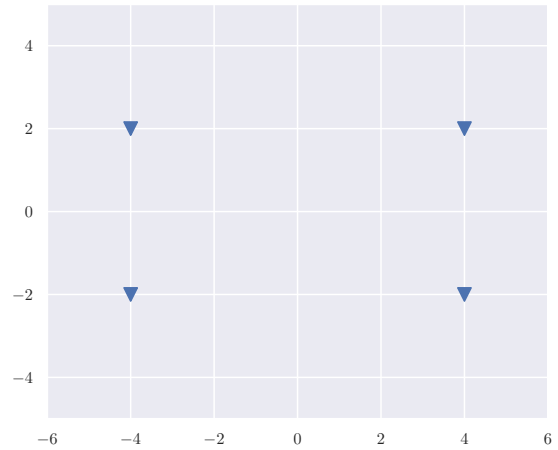
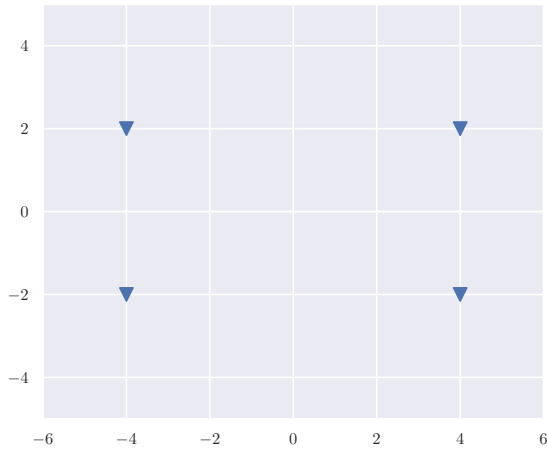


Problem 9 GMMs & EM (3 credits)

This problem is about Gaussian mixture models (GMM) and fitting them to data with the expectation maximization (EM) algorithm.

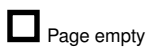
In this problem, the observed data are depicted as downwards pointing triangles.

- 0 ☐
- 1 ☐
- a) In the figures below, sketch two possible optima that we can achieve when fitting a 2-component GMM to this data with EM. When you sketch components of a GMM, mark the component center with a cross (x) and indicate the shape of the component covariance matrix with an ellipse.

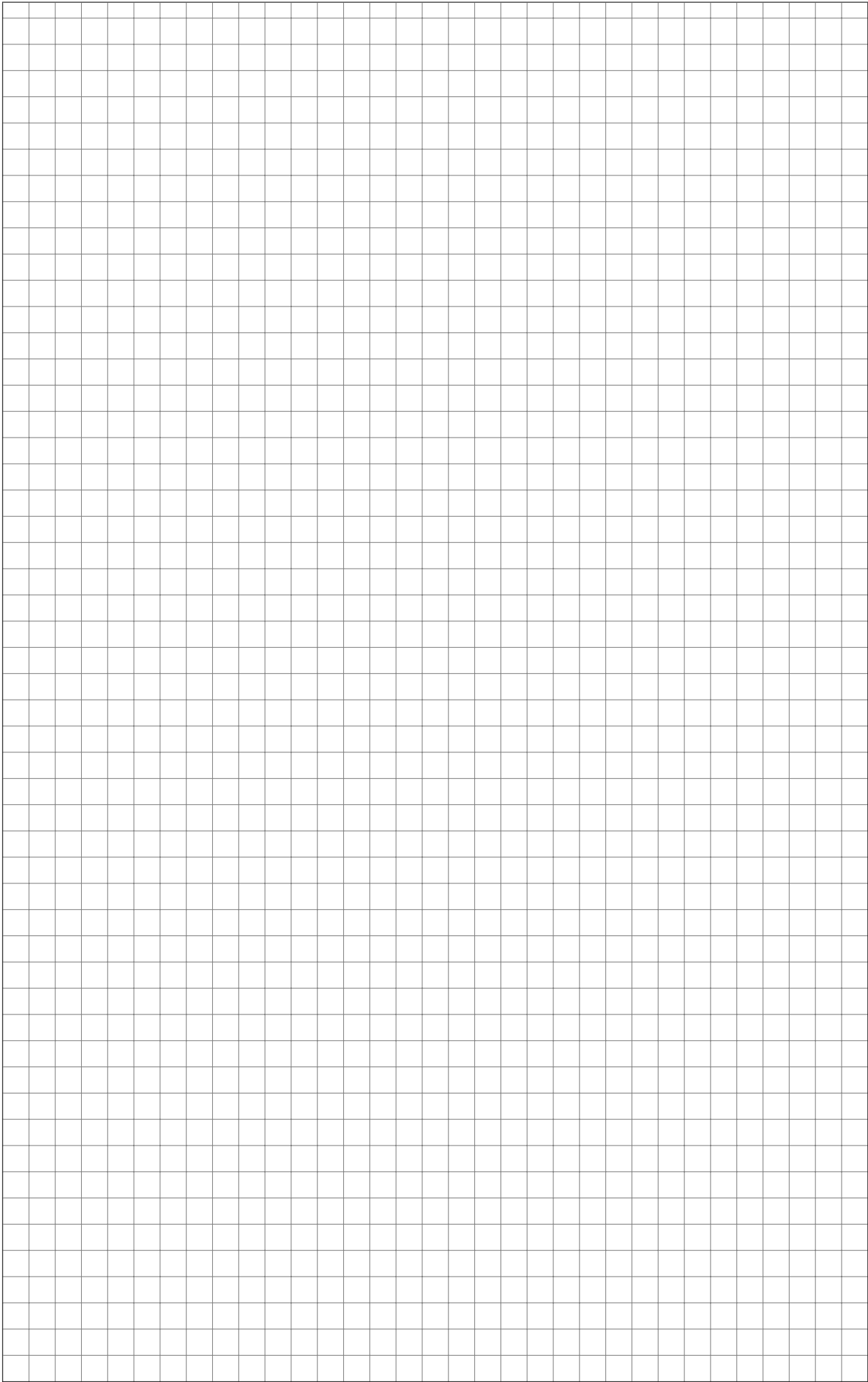


- 0 ☐
- 1 ☐
- 2 ☐
- b) Consider the three 1-dimensional data points $\mathcal{D} = \{0, 1, 3\}$. We want to run K-means++ to initialize $K = 2$ centroids. What is the probability that the two centroids are initialized in different “clusters”, i.e. one centroid in the subset $\{0, 1\} \subset \mathcal{D}$ and one centroid in $\{3\} \subset \mathcal{D}$? **Justify your answer!**





CIT-IN2064-01-20260212
4794-01-0008-E0832-15



4794-01-0008-E0832-16



4794-01-0008-E0832-16



4794-01-0008-E0832-16

